



Answers & Solutions:

1. C. I_L and I_C are 180° out of phase.

2. D. Reactive power.

3. A. resistive

4. D. All of the above.

5. B. 14.14 A

Solution:

$$\text{rms value} = 20A(0.707)$$

$$\text{rms value} = 14.14 \text{ A} \rightarrow \text{Ans.}$$

6. A. 14.32 (lagging)

Solution:

$$\phi = 0.25 \text{ radians lagging}$$

$$0.25 \text{ rads} = 0.25 \times \frac{180^\circ}{\pi}$$

$$= 14.32^\circ \text{ (lagging)} \rightarrow \text{Ans.}$$

7. A. I_L lags I_R by 90° .

8. D. 1.257 A

Solution:

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 100 \times 10 \times 10^{-6}}$$

$$X_C = 159.155 \Omega$$

$$I = \frac{V}{X_C} = \frac{200}{159.155} = 1.257 \text{ A} \rightarrow \text{Ans.}$$

9. B. 1.

10. A. current leads the voltage by 45°

11. A. V_L leads i_L by 90° .

12. B. Active power increases

13. A. 397.9 ohms

Solution:

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 20 \times 20 \times 10^{-6}}$$

$$X_C = 397.9 \Omega \rightarrow \text{Ans.}$$

14. C. reactance

15. A. X_L and X_C are 180° out of phase.

16. A. current leads the voltage by 45°

17. C. V_C lags V_R by 90°

18. A. 628 Ω ; 0.318 A

Solution:

$$X_L = 2\pi fL = 2\pi(2 \times 10^3)(50 \times 10^{-3})$$

$$X_L = 628 \Omega \rightarrow \text{Ans.}$$

$$I = \frac{V}{X_L} = \frac{200}{628}$$

$$I = 0.318 \text{ A} \rightarrow \text{Ans.}$$

19. C. capacitive

20. D. It allows independent operations of loads

21. C. Root mean square (RMS).

22. C. Z_T decreases.

23. A. $V_s = 97.39 \sin(300\pi t + 0.620) \text{ V}$

Solution:

Supply voltage,

$$V_s = v_1 + v_2 + v_3$$

$$V_s = 30 \sin\left(300\pi t - \frac{\pi}{6}\right) + 40 \sin\left(300\pi t + \frac{\pi}{4}\right)$$

$$+ 50 \sin\left(300\pi t + \frac{\pi}{3}\right)$$

Using complex number

$$V_s = 30 \angle -30^\circ + 40 \angle 45^\circ + 50 \angle 60^\circ$$

$$V_s = 79.265 + j56.586$$

$$V_s = 97.39 \angle 35.52^\circ \text{ V or}$$

$$V_s = 97.39 \angle 0.620 \text{ V}$$

$\therefore$

$$V_s = 97.39 \sin(300\pi t + 0.620) \text{ V} \rightarrow \text{Ans.}$$

24. A. Apparent power.

25. D. $150 - j100$

Solution:

$$Z = R + jX = 150 - j100 \rightarrow \text{Ans.}$$

where:

$$X_L = 100 \Omega$$

$$X_C = 200 \Omega$$

$$X = X_L - X_C = 100 \Omega - 200 \Omega$$

$$X = -100 \Omega$$

$$R = 150 \Omega$$

26. B. V_T lags I_T by 90° .

27. B. 500Ω of capacitive reactance

28. B. applies only to sine waves.

29. B. reciprocal of reactance.

30. B. 18Ω

Solution:

$$Z = \sqrt{R^2 + X^2} = \sqrt{(6 \Omega)^2 + (17 \Omega)^2}$$

$$Z = 18 \Omega \rightarrow \text{Ans.}$$

where:

$$X = X_L - X_C = 17 \Omega - 0 \Omega = 17 \Omega$$

$$R = 6 \Omega$$

$$X_L = 17 \Omega$$

$$X_C = 0 \Omega$$

31. C. Z_{EQ} decreases.

32. C. $19.1 - j11.1$

Solution:

$$i(t) = 10 \cos(\omega t - 30^\circ)$$

$$= 10 \sin(\omega t + 60^\circ)$$

$$v(t) = 220 \sin(\omega t + 30^\circ)$$

Use Mode 2 (CMPLX)

$$Z = \frac{V}{I} = \frac{220 \angle 30^\circ}{10 \angle 60^\circ}$$

$$Z = 19.1 - j11.1 \rightarrow \text{Ans.}$$

33. C. I_C leads I_R by 90° .

34. C. inductive



35. B. $1 - j1.5$

Solution:

$$B_L = \frac{1}{X_L} = \frac{1}{2\pi fL} = 1 \text{ S}$$

$$B_C = \frac{1}{X_C} = 2\pi fC = 1 \text{ S}$$

If f is halved

$$B_L = \frac{1 \text{ S}}{\frac{1}{2}} = 2 \text{ S}$$

$$B_C = \frac{1 \text{ S}}{2} = 0.5 \text{ S}$$

$$G = \frac{1}{R} = \frac{1}{1 \Omega} = 1 \text{ S}$$

$$B = B_C - B_L = 0.5 \text{ S} - 2 \text{ S} = -1.5$$

$$Y = G + jB = 1 - j1.5 \rightarrow \text{Ans.}$$

36. C. 250 mA of capacitive current.

37. A. 15.92 μF ; 0.25 A

Solution:

$$X_C = \frac{1}{2\pi fC}$$

$$C = \frac{1}{2\pi fX_C} = \frac{1}{2\pi(25)(400)}$$

$$C = 15.92 \mu\text{F} \rightarrow \text{Ans.}$$

$$I = \frac{V}{X_L} = \frac{100}{400} = 0.25 \text{ A} \rightarrow \text{Ans.}$$

38. B. It alternates between magnetic and electric fields and is not dissipated.

39. D. Multiply apparent power times the power factor.

40. A. V_A and I_R are in phase.

41. A. 40 ohms

Solution:

$$X_L = 2\pi fL = \omega L$$

$$X_L = (240)(100 \times 10^{-3})$$

$$X_L = 24 \Omega$$

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{32^2 + 24^2}$$

$$Z = 40 \Omega \rightarrow \text{Ans.}$$

42. B. A wave whose amplitude at any given instant can be represented by the projection of a point on a wheel rotating at a uniform speed.

43. D. 37.7 V

Solution:

$$X_L = 2\pi fL = 2\pi(1 \times 10^3)(120 \times 10^{-3})$$

$$X_L = 753.982 \Omega$$

$$V_L = IX_L = (50 \times 10^{-3})(753.982)$$

$$V_L = 37.7 \text{ V} \rightarrow \text{Ans.}$$

44. C. Sawtooth wave.

45. A. Square.

46. D. Both A and C.

47. D. 360 degrees.

48. C. V_R and I are in phase.

49. D. A wave that abruptly changes back and forth that abruptly changes back and forth between two voltage levels and stays at these levels for equal amounts of time.

50. A. Sawtooth wave.

END